

Multivariate Asynchronous Shapelets for Imbalanced Car Crash Predictions

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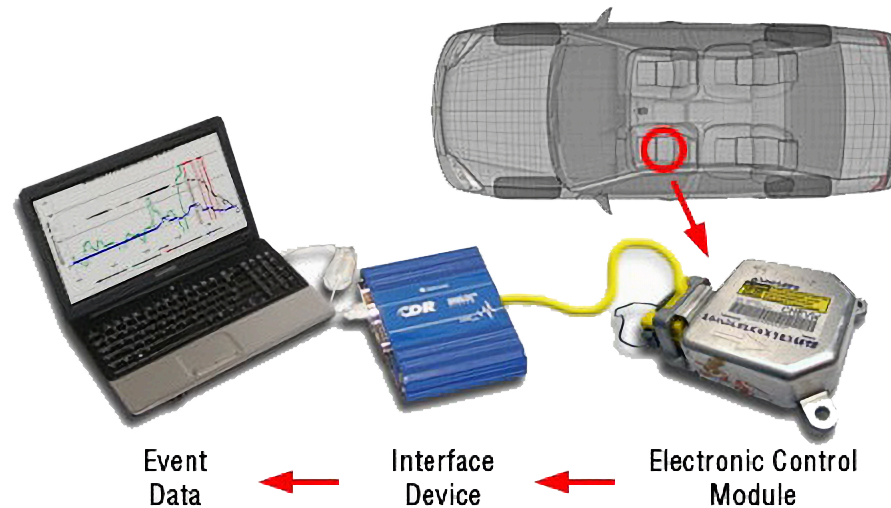
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DS 2024



Crash Data Recorders

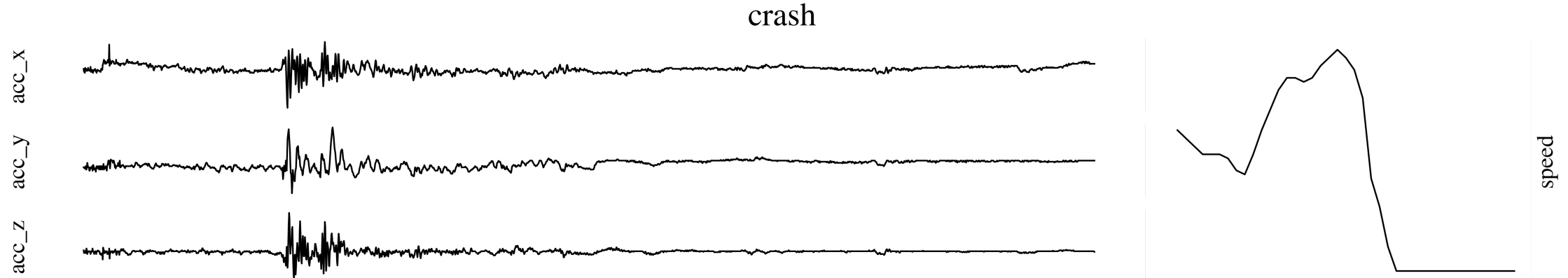
Crash Data Recorders (CDRs) can retrieve car's event data which monitor safety measures and record impact speeds.



We collaborate with Assicurazioni Generali to detect car crashes.

Generali's Dataset

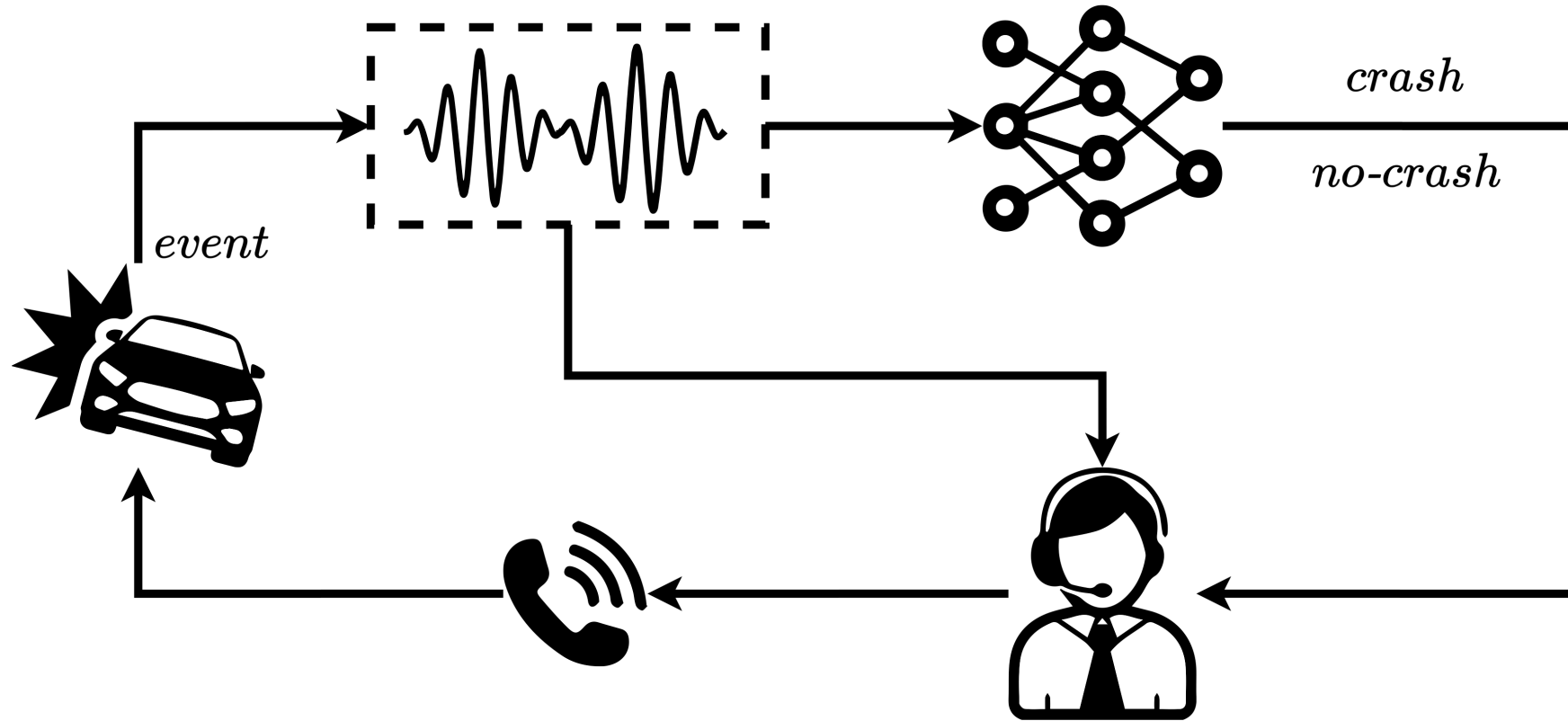
Generali's dataset is composed by **multivariate** time series containing 4 signals (acceleration on x,y,z axis and speed).



It is a challenging **binary classification dataset**, big, highly imbalanced, with classes: *Crash* ($\sim 1\%$) and *No-Crash* ($\sim 99\%$).

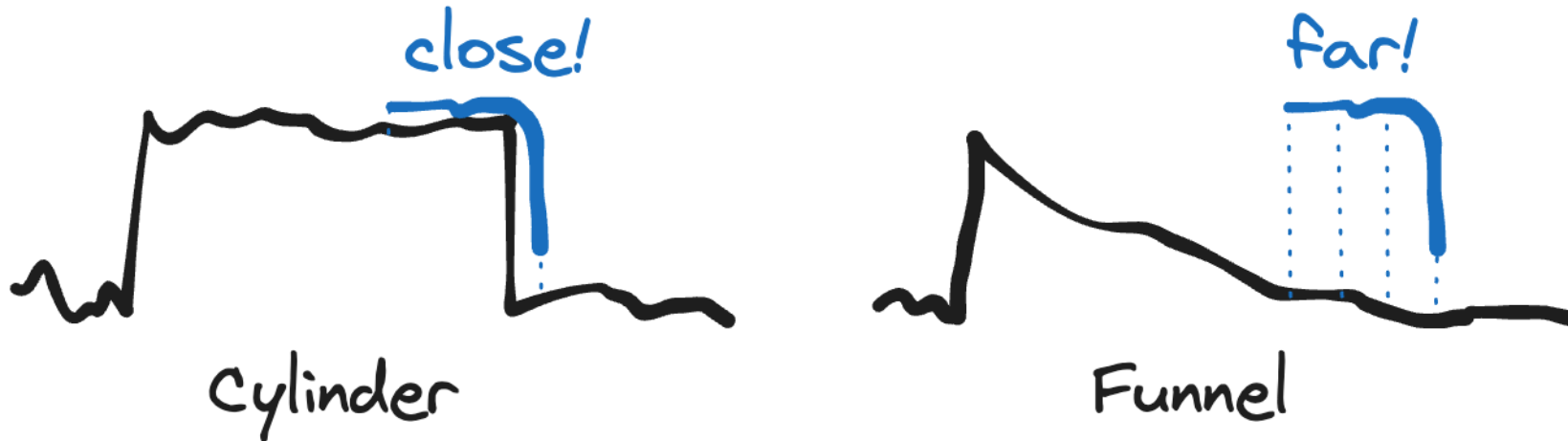
Generali's Workflow

Generali has a customer assistance workflow with a human-in-the-loop



Shapelets

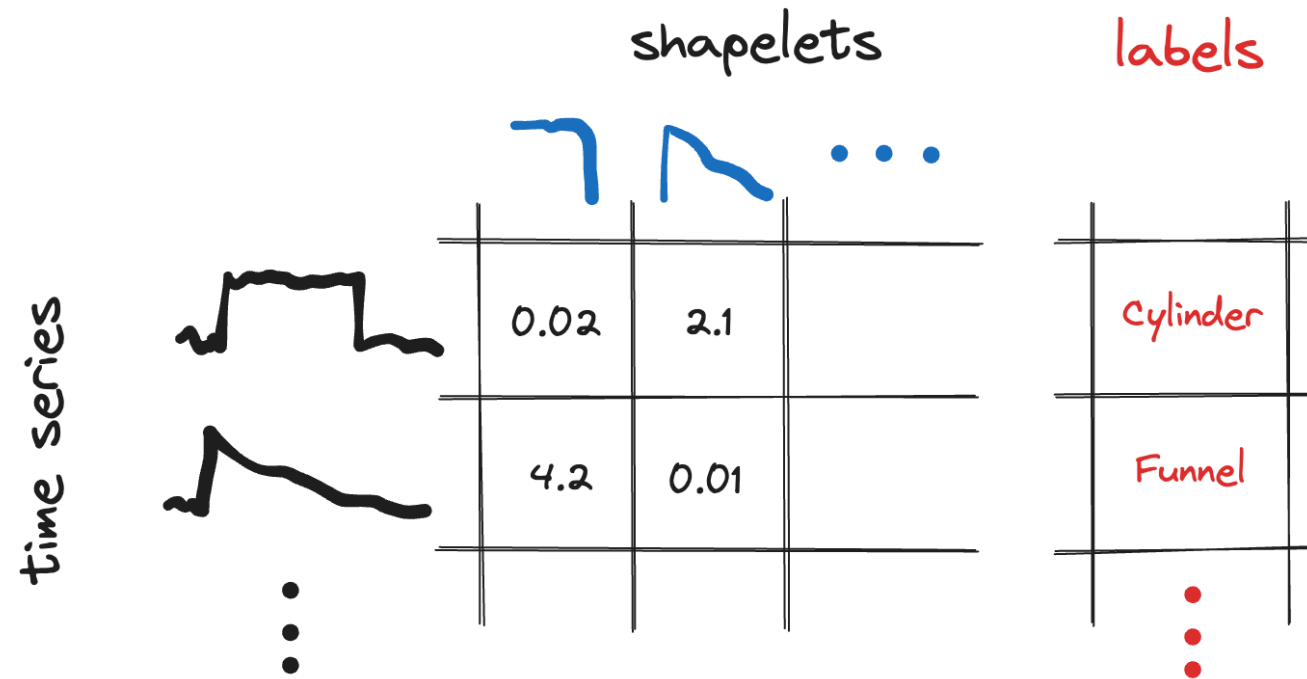
Highly representative and **discriminative** time series subsequences for a particular class in a time series dataset.



They can be extracted in many supervised and unsupervised ways.

Shapelet Transform

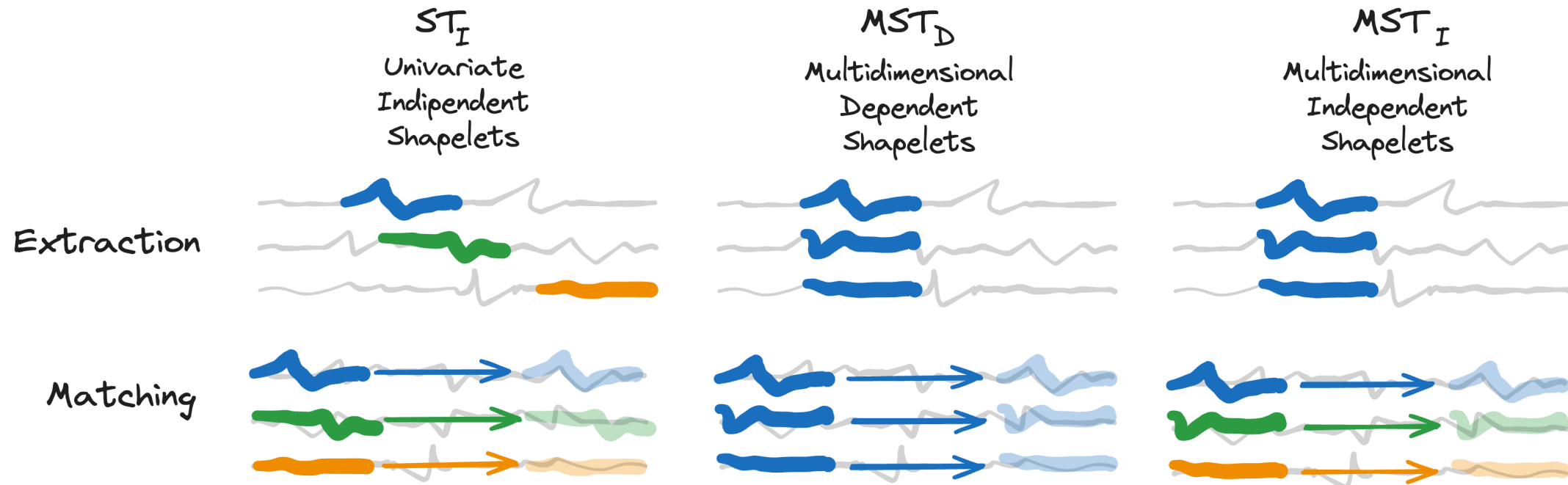
Represent the time series dataset in a **tabular form**, as the **minimum sliding-window distance** between each shapelet and each time series.



Even for multivariate data, usually shapelets are **univariate**.

Multivariate Shapelets

Multivariate shapelets are scarcely addressed in the literature.

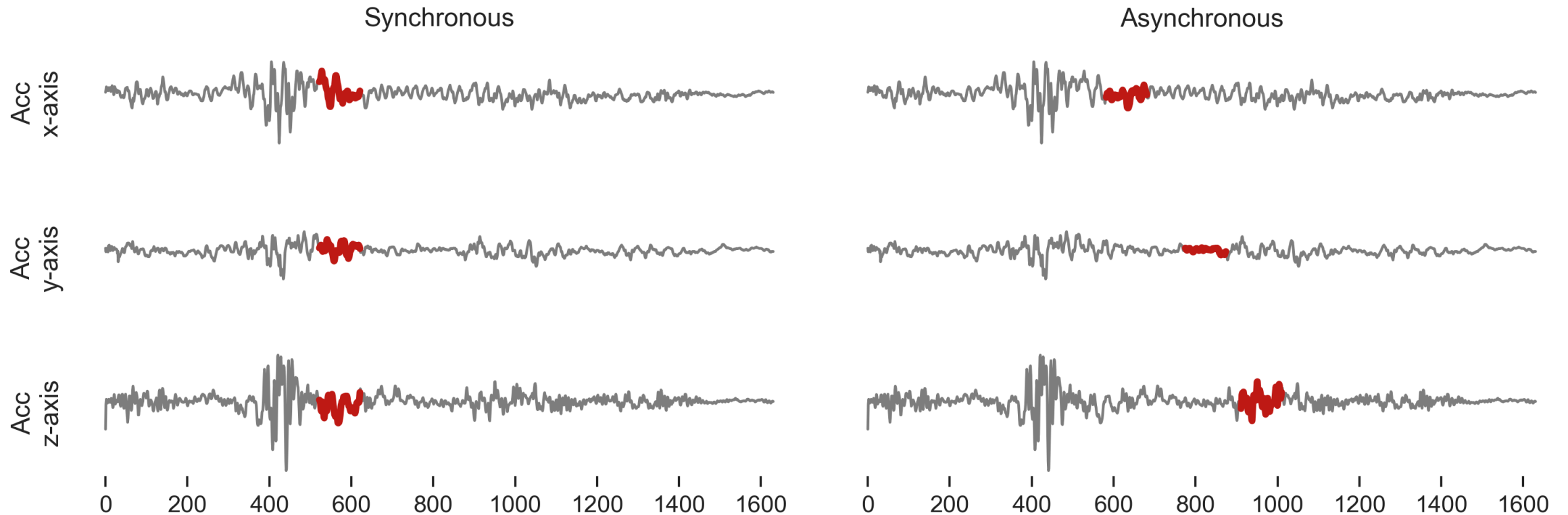


There are several limitations to these approaches:

- ST_I shapelets are univariate;
- MST_D , MST_I only extract aligned/synchronous shapelets;
- poor performance on **imbalanced** datasets;
- **interpretability** is assumed but **rarely explored**;
- computational complexity.

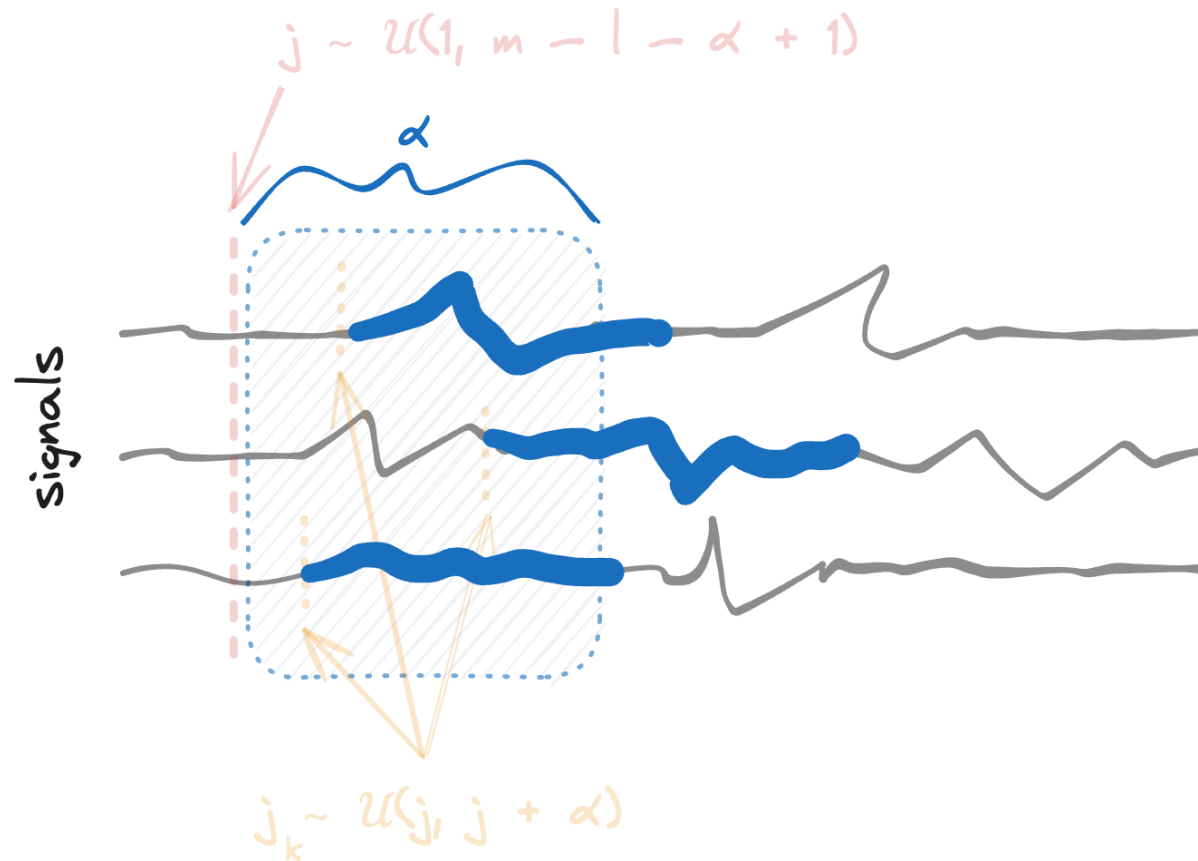
Multivariate Asynchronous Shapelets (MARS)

We want to control the **asynchronicity** of the shapelets.



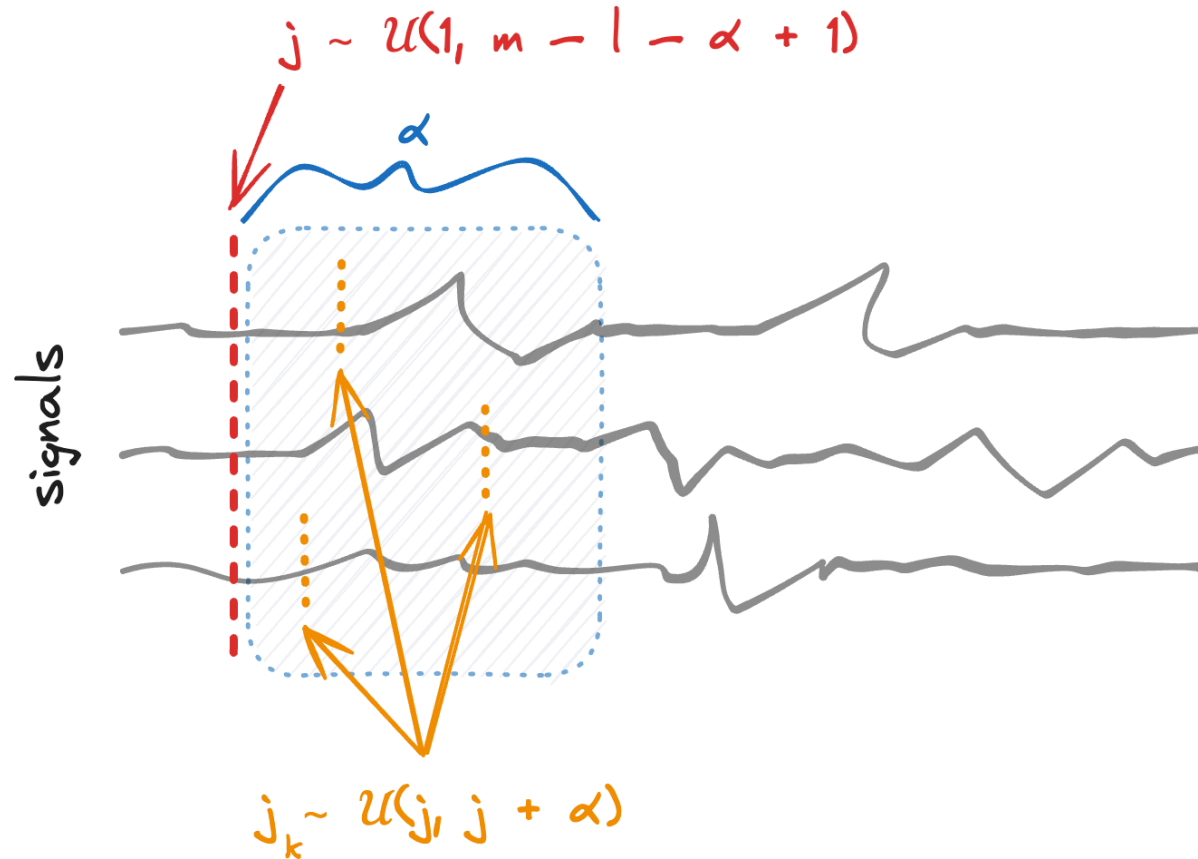
Shapelet Extraction

α defines how much the multivariate shapelet can be misaligned.



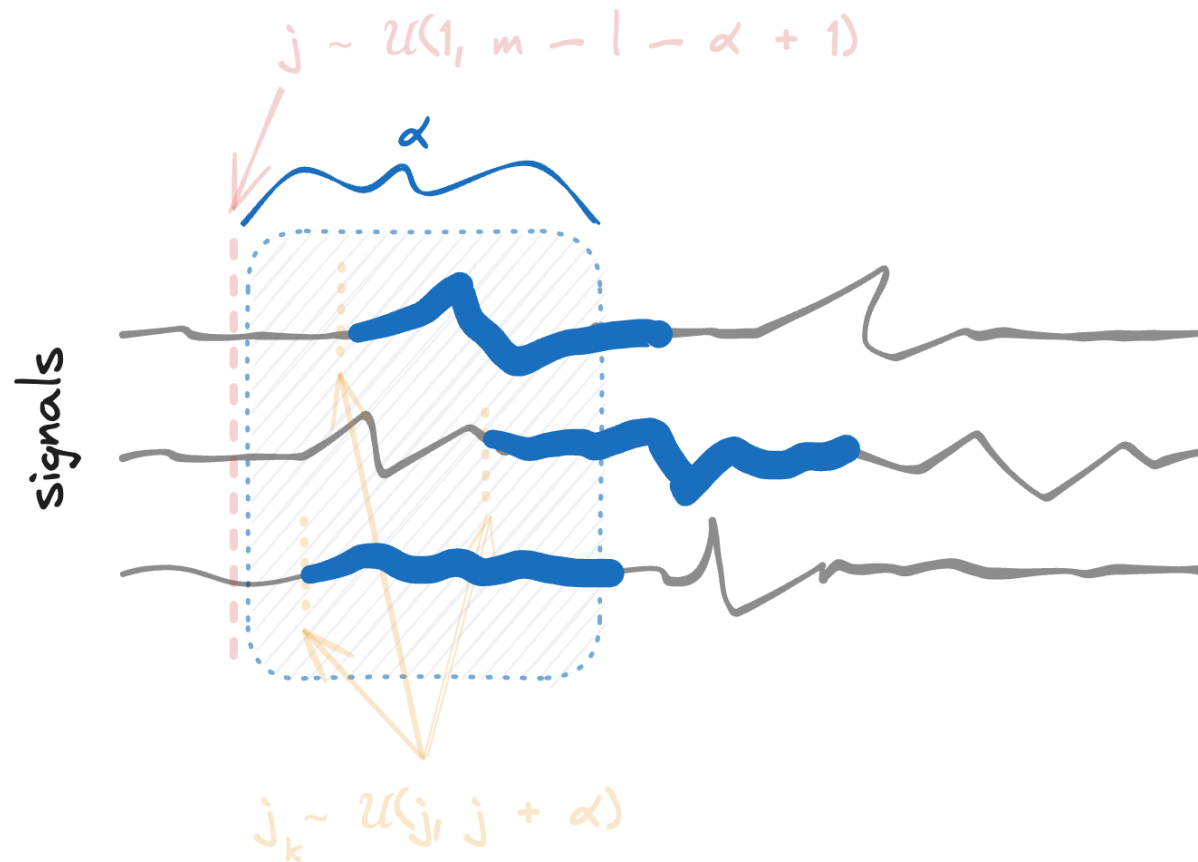
Shapelet Extraction

To enforce this we perform two index samplings: **global** and **specific**.



Shapelet Extraction

This ensures that the starting indexes are within α timesteps.



Shapelet Extraction for Imbalanced Datasets

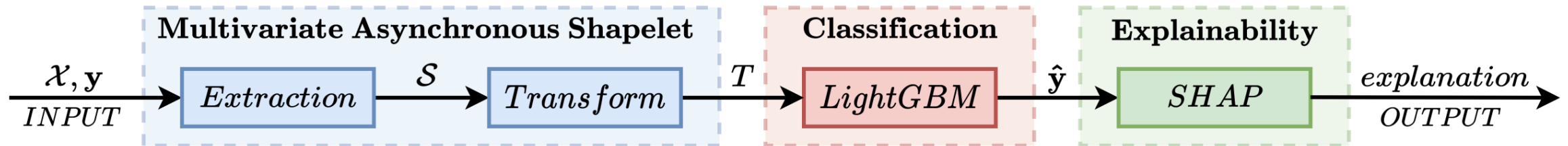
The **brute force** extraction of shapelets is **not feasible**, therefore, we randomly extract h shapelets from the TSC dataset.

Specifically we extract $\lfloor \frac{h}{2} \rfloor$ shapelets from time series belonging to **each of the 2 classes**.

This ensures that, if h is sufficiently high, both *Crash* and *No-Crash* instances will be represented by some shapelets.

MARS Pipeline

The Multivariate Asynchronous Shapelet extraction is used in a classification pipeline to **predict and explain** car crashes.



We first benchmark MARS on Generali's dataset:

- we find the best MARS configuration;
- we compare it to the state-of-the-art.

Given the highly imbalanced dataset we need to use appropriate metrics* such as: Geometric Mean ($\sqrt{sensitivity \times specificity}$), F1-score, and Gini coefficient ($2 \times AUC - 1$).

*Japkowicz, Nathalie. "Assessment metrics for imbalanced learning." Imbalanced learning: Foundations, algorithms, and applications (2013): 187-206.

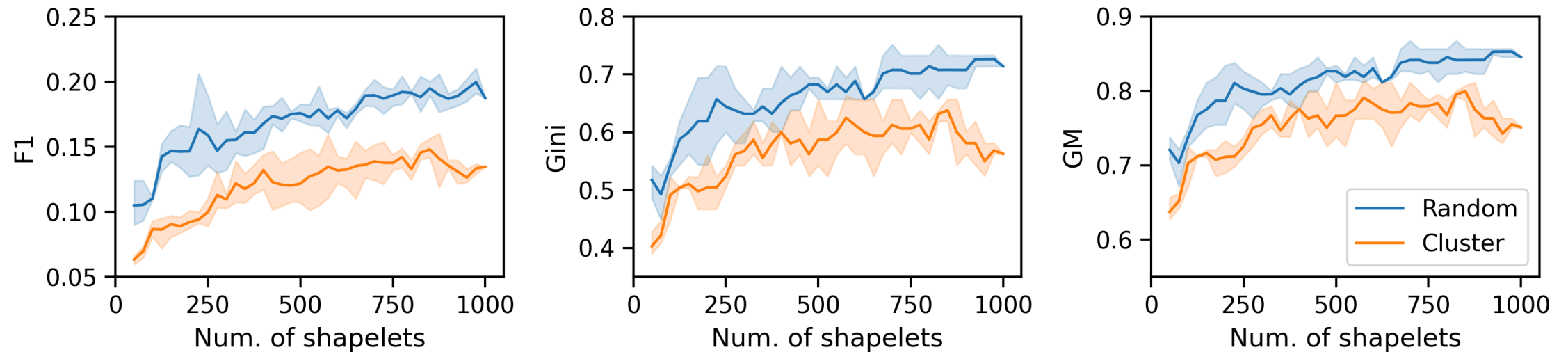
MARS Configuration (Shapelet Length and Asynchronicity)

shapelet type	l	α	$multi$	$f1$	gm	$gini$
short, sync	8-20	0	✓	0.11	0.74	0.54
short, low async	8-20	20	✓	0.07	0.63	0.39
short, high async	8-20	max	✓	0.14	<u>0.80</u>	<u>0.64</u>
long, sync	20-40	0	✓	0.09	0.66	0.43
long, low async	20-40	20	✓	0.08	0.70	0.48
long, high async	20-40	max	✓	<u>0.17</u>	0.79	0.62
x-axis only	8-20	✗	✗	0.11	0.72	0.52
speed-axis only	8-20	✗	✗	0.04	0.47	0.22

More asynchronous shapelets work better, length is less relevant.

MARS Configuration (Number of Shapelets)

We compare **random** extraction against **cluster**-based extraction increasing the number of extracted shapelets.



~ 1000 randomly extracted shapelets seem to be enough.

State-Of-The-Art Comparison

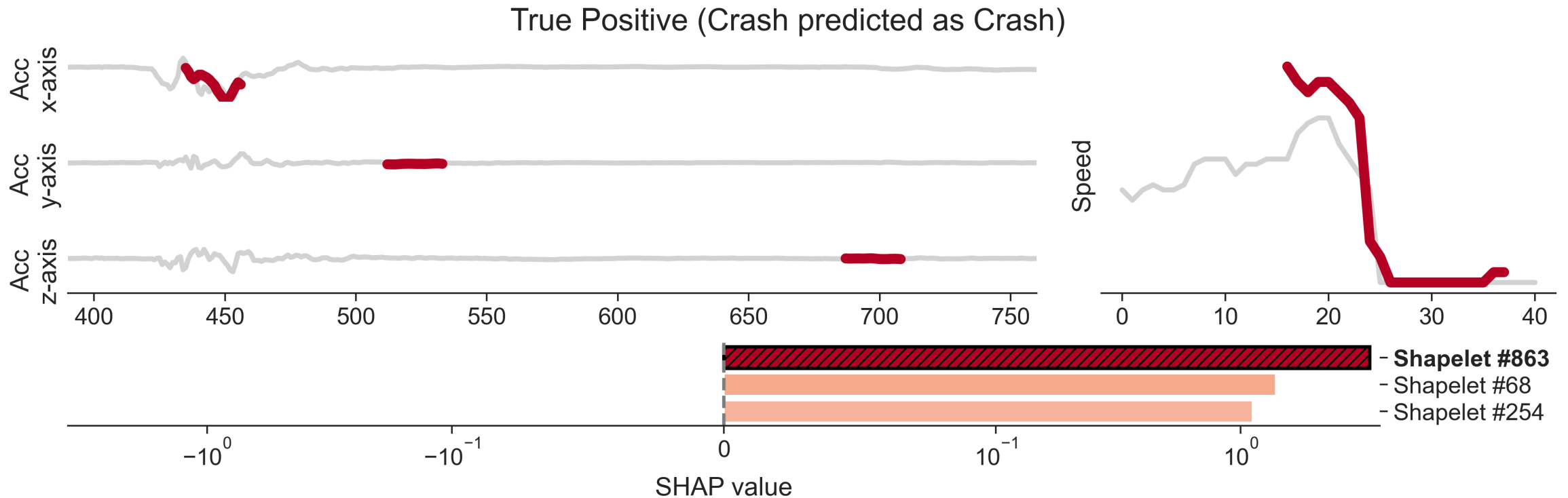
We compare MARS against the state-of-the-art **classifiers** and **anomaly detection** algorithms.

	MARS	GEN _L	GEN _H	TSF	ROCKET	ROCKAD	XGB	LGBM
<i>f1</i>	0.19	0.17	0.31	0.18	0.28	0.01	0.14	0.12
<i>gini</i>	0.71	0.66	0.47	0.64	0.53	0.28	0.58	0.66
<i>gm</i>	0.84	0.81	0.69	0.80	0.73	0.56	0.76	0.81
<i>tp</i>	38	35	25	34	28	18	31	35
<i>fp</i>	315	330	84	283	121	6122	349	514
<i>fn</i>	15	18	28	19	25	35	22	18
$\frac{time_{tr}}{time_{in}}$	17 h	-*	-*	3 h	1 h	5 h	5 min	1 min
	0.82 s	0.09 s	0.09 s	0.14 s	0.07 s	0.01 s	35 μ s	30 μs

* Data unavailable due to training on Generali's system.

Explaining Car Crashes (True Positive)

MARS returns an explanation in terms of **multivariate shapelet contribution** (SHAP values) toward either the class *Crash* or *No-Crash*.



Limitations and Future Works

There are a few limitations:

- the **training time** is still slow;
- the **asynchronicity limit** is only applied in the extraction phase, **not in the matching phase**;
- explanations are in terms of **feature importance only**.

Explanations could be improved by making them **global** and **more expressive**, e.g., decision trees.

THANK YOU FOR THE ATTENTION!

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 <https://github.com/bianchimario/MARS>

Bonus Slide - Multivariate Shapelets

Multivariate shapelets are scarcely addressed in the literature.

- ST_I - independent shapelets: extracted **independently** from each signal and assessed individually (basically univariate).
- MST_D - multidimensional dependent shapelets: **maintain phase alignment** across all channels for extraction and matching phases.
- MST_I - multidimensional independent shapelets: MST_D shapelet extraction, **signal-independent matching**.

Bonus Slide - Shapelet Extraction

We define, α , the asynchronicity limit parameter which defines how much misalignment can exist between shapelets of each channel:

$$\max_{k,k'} |\dot{j}_k - \dot{j}_{k'}| \leq \alpha.$$

The maximum distance between the starting indices, $\dot{j}$, of two channels k, k' can be at most α .

Bonus Slide - Shapelet Extraction

First we sample a time series generic starting index:

$$j \sim \mathcal{U}(1, m - l - \alpha + 1).$$

Then, we sample a signal-specific starting index:

$$j_k \sim \mathcal{U}(j, j + \alpha).$$

Bonus Slide - Shapelet Matching

Shapelet matching is performed by finding the minimum distance between each signal and the respective shapelet dimension:

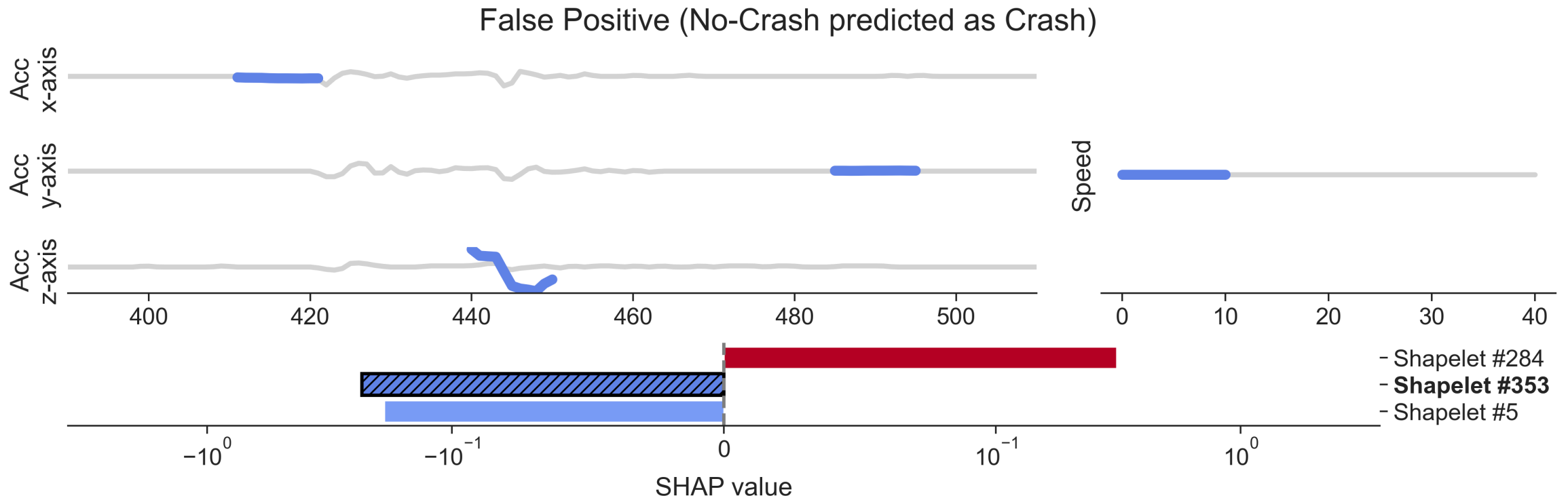
$$D_k(X, S) = \min_{j \in \{1, \dots, m-l+1\}} \sqrt{\sum_{i=0}^{l-1} (x_{k,j+i} - s_{k,i})^2}.$$

These distances are then summed:

$$D(X, S) = \sum_{k=1}^d D_k(X, S).$$

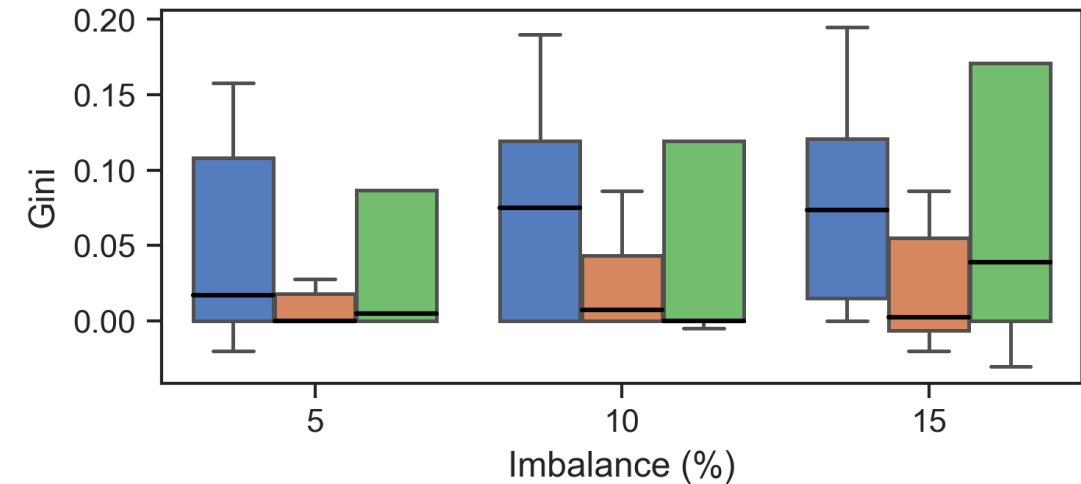
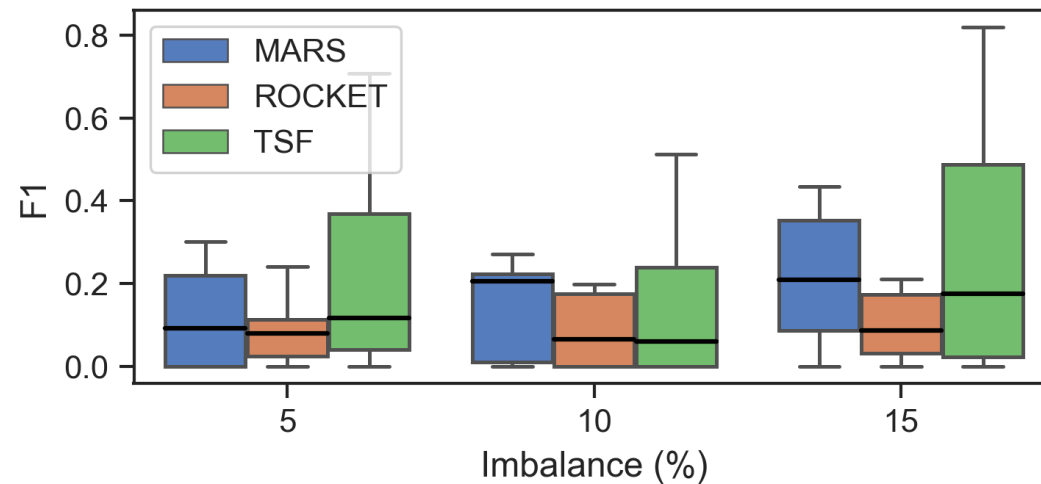
Bonus Slide - Explaining Car Crashes (False Positive)

MARS returns an explanation in terms of **multivariate shapelet contribution** (SHAP values) toward either the class *Crash* or *No-Crash*.



Bonus Slide - UEA Repository Benchmarks

We also benchmark MARS on standard, synthetically imbalanced, multivariate time series classification datasets.



Bonus Slide - UEA Repository Benchmarks

	<i>f1</i>			<i>gini</i>			<i>gm</i>		
	5%	10%	15%	5%	10%	15%	5%	10%	15%
MARS	.12±.12	.22±.28	.26±.25	.05±.07	.15±.24	.12±.17	.05±.15	.15±.31	.11±.26
ROCKET	.08±.08	.16±.27	.18±.29	.02±.05	.09±.20	-.03±.50	.07±.14	.10±.28	.13±.30
TSF	.22±.25	.20±.30	.28±.31	.10±.19	.12±.22	.16±.27	.09±.26	.10±.27	.14±.29